

# Marketing Planning Assistant Agent

Course Name: Agentic AI

***Institution Name:*** Medicaps University – Datagami Skill Based Course

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*Academic Year:* 2025-2026

# Problem Statement & Objectives:-

## 1. Problem Statement

In the modern digital economy, developing an effective marketing plan is essential for business success. Marketing strategy creation involves multiple complex tasks such as competitor analysis, keyword research, budget planning, and campaign scheduling. These activities require expertise, time, and access to specialized tools, which many startups, small businesses, and individuals often lack.

Manual marketing planning can lead to unstructured strategies, inefficient budget allocation, and suboptimal targeting decisions. Despite advancements in Artificial Intelligence and Large Language Models (LLMs), there is still a gap in systems that can intelligently transform high-level marketing goals into structured and actionable execution plans.

Therefore, there is a need for an AI-powered solution that can assist users by automating marketing plan generation, improving strategic structure, and simplifying decision-making. This project addresses this need by developing an **AI-Powered Marketing Planning Assistant Agent** that leverages LLM-based reasoning, modular tools, and an interactive Streamlit interface.

## 2. Project Objectives

The primary objectives of this project are:

- To design and develop an AI-powered Marketing Planning Assistant Agent.
- To automate the generation of structured marketing execution plans.
- To leverage Large Language Models (LLMs) for intelligent reasoning and strategy formulation.
- To integrate modular tools for keyword research, budgeting, scheduling, and ad analysis.
- To develop an interactive and user-friendly graphical interface using Streamlit.
- To enable export of marketing plans in PDF format.
- To implement persistent data storage using MySQL database.
- To enhance user experience through features like plan history, search, deletion, and dark mode

### 3. Scope of the Project

This project focuses on developing an AI-powered Marketing Planning Assistant Agent that generates structured marketing plans based on high-level user goals. The system utilizes Large Language Models (LLMs) through LangChain to provide intelligent strategy recommendations.

The scope includes:

- User interaction via Streamlit-based graphical interface.
- Automated marketing plan generation.
- Integration of tools for keyword research, budgeting, scheduling, and ad analysis.
- Features such as plan history, search, loading, and deletion.
- PDF export functionality.
- Persistent storage using MySQL database

## Proposed Solution:-

An AI-driven Marketing Planning Assistant Agent was developed that accepts a high-level marketing goal from the user and produces a structured execution plan. The system uses Large Language Models (LLMs) integrated through LangChain, supported by modular tools for keyword research, budgeting, scheduling, and ad analysis.

A Streamlit-based graphical interface enables easy interaction, while MySQL ensures persistent storage of plans.

### 1. Key features

- AI-powered marketing plan generation
- Structured execution workflow
- Tool-integrated reasoning
- Streamlit interactive GUI
- Dark mode UI
- Plan history & retrieval
- PDF download option
- MySQL database integration
- Delete & search plans

## 2. Overall Architecture / Workflow

The AI-Powered Marketing Planning Assistant Agent follows a modular and layered architecture designed to ensure scalability, flexibility, and ease of interaction. The system integrates a user-friendly interface, core AI logic, auxiliary tools, and a persistent database layer.

### Workflow Overview:

#### 1 User Interaction

The user enters a high-level marketing goal through the Streamlit graphical interface.

#### 2 Input Processing

The input goal is passed to the MarketingAgent for analysis and reasoning.

#### 3 LLM Invocation

The MarketingAgent invokes the Large Language Model (Google Generative AI) via LangChain to interpret the goal and generate a structured marketing plan.

#### 4 Tool Integration (Optional)

When required, the agent utilizes integrated tools such as Keyword Tool, Budget Tool, Scheduler Tool, and Ad Library Tool to enrich the response.

#### 5 Plan Generation

A structured marketing execution plan is generated based on AI reasoning and tool outputs.

#### 6 Output Presentation

The generated plan is displayed within the Streamlit interface.

#### 7 PDF Export

The system enables users to download the plan as a PDF using ReportLab.

#### 8 Database Storage

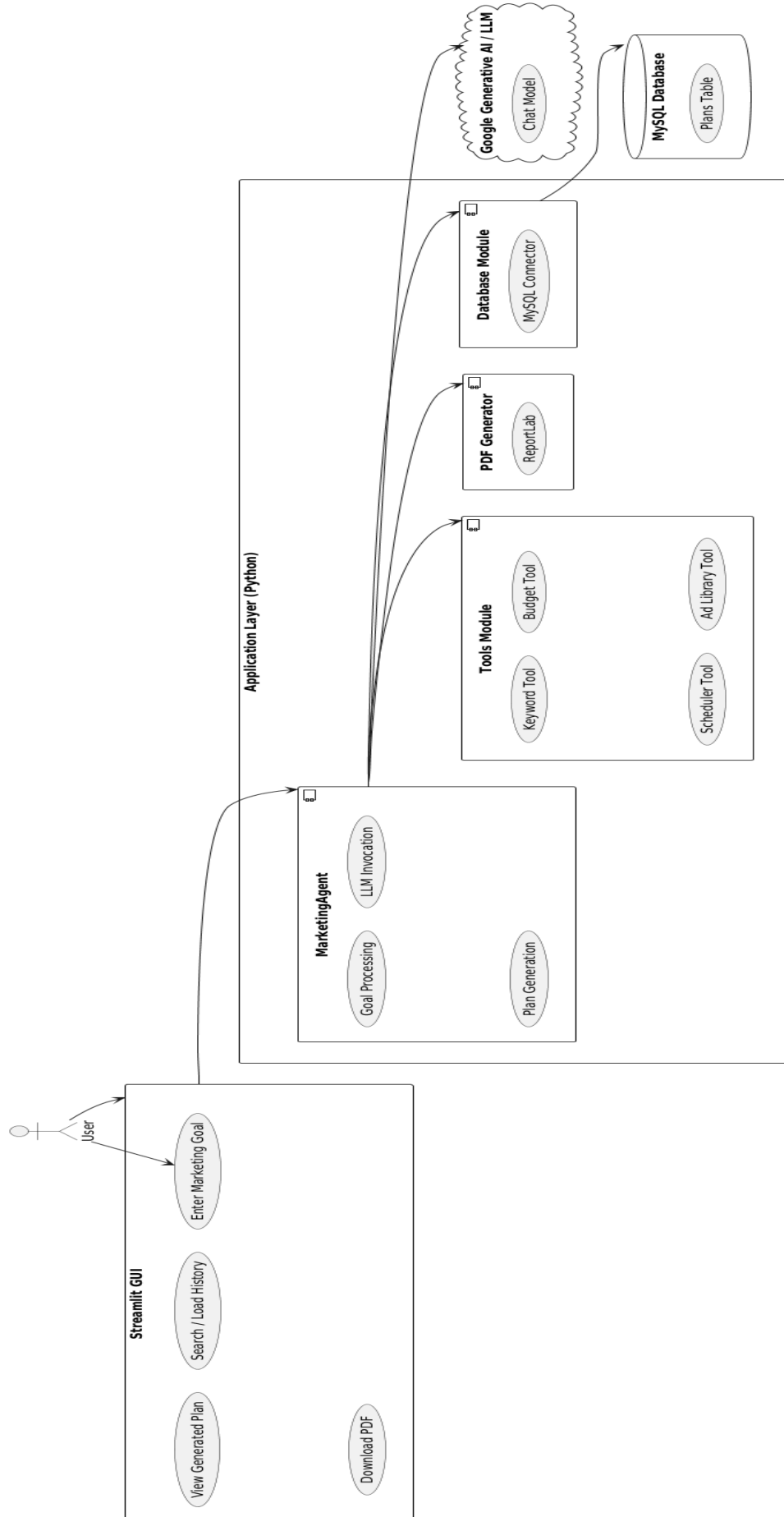
The plan, along with its metadata, is stored in the MySQL database for persistence.

#### 9 History & Retrieval

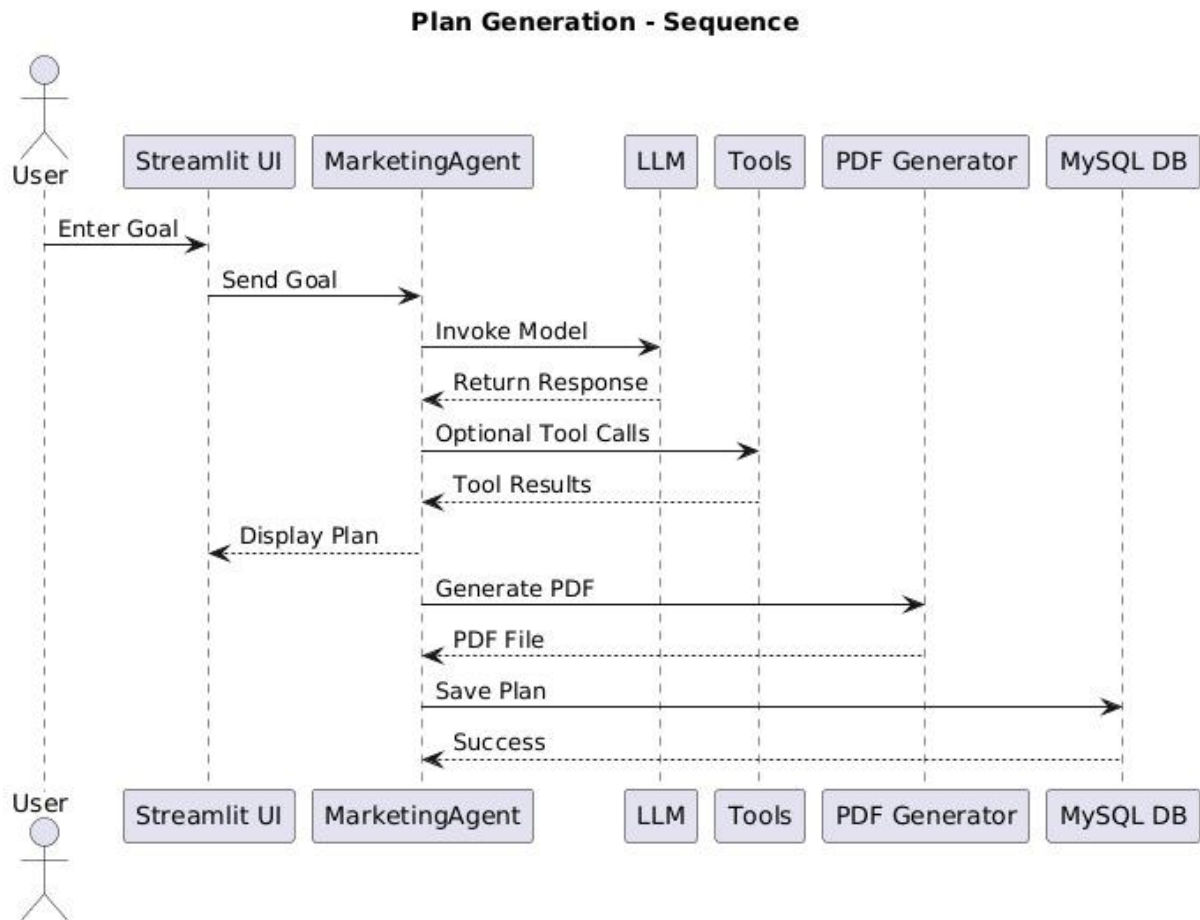
Users can search, load, and delete previously generated plans via the sidebar.

## 1. System Architecture Diagram

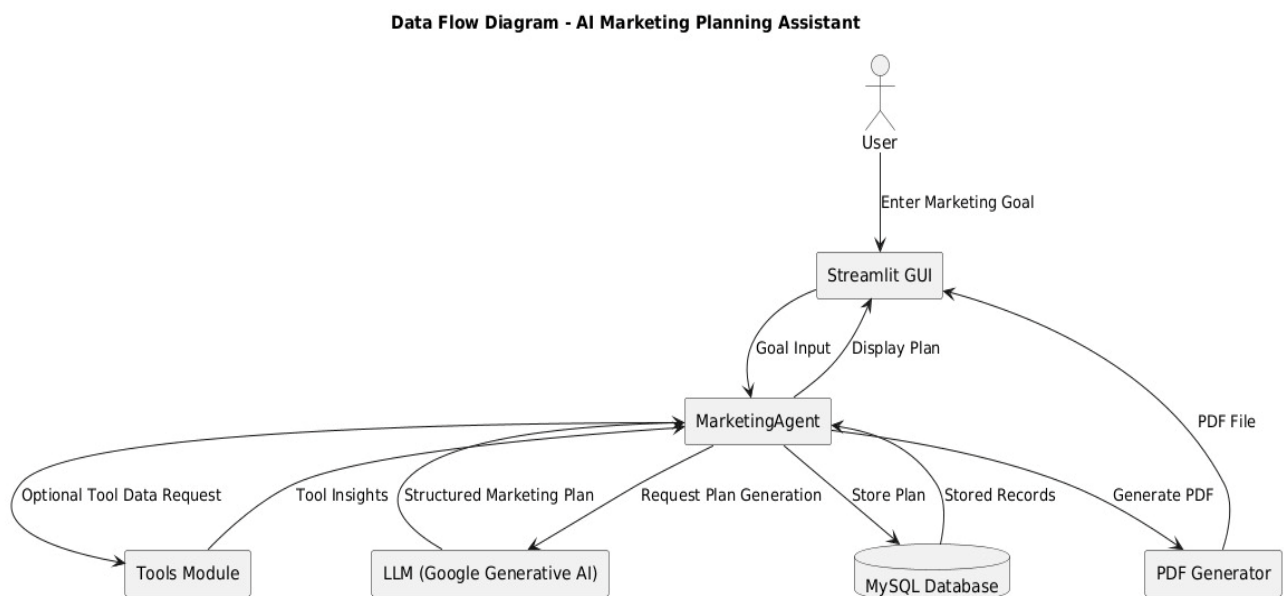
# AI-Powered Marketing Planning Assistant Agent - Architecture



## 2. Sequence Diagram



## 3. Data Flow Diagram



### 3.Tools & Technologies Used

The development of the AI-Powered Marketing Planning Assistant Agent involved the use of the following tools and technologies:

#### Programming Language :-

- **Python** – Python is the primary programming language used to build the Marketing Planning Assistant Agent. It was chosen due to its simplicity, readability, and strong ecosystem for AI and automation. Python enables rapid development and easy integration with AI frameworks, APIs, and data processing tools.

#### Frameworks & Libraries :-

- **Streamlit** – Used to develop the interactive graphical user interface (GUI).
- **LangChain** – Enabled integration of Large Language Models (LLMs) and tool-based reasoning.
- **Pydantic** – Used for data validation and structured plan modeling.
- **ReportLab** – Used for generating downloadable PDF reports.

#### Artificial Intelligence :-

- **Google Generative AI (LLM)** – Provided intelligent reasoning and marketing plan generation.

#### Database :-

- **MySQL** – Used for persistent storage of generated marketing plans.
- **MySQL Connector (Python)** – Enabled communication between Python application and MySQL database.

#### Environment Management :-

- **python-dotenv** – Used for secure handling of API keys and database credentials.

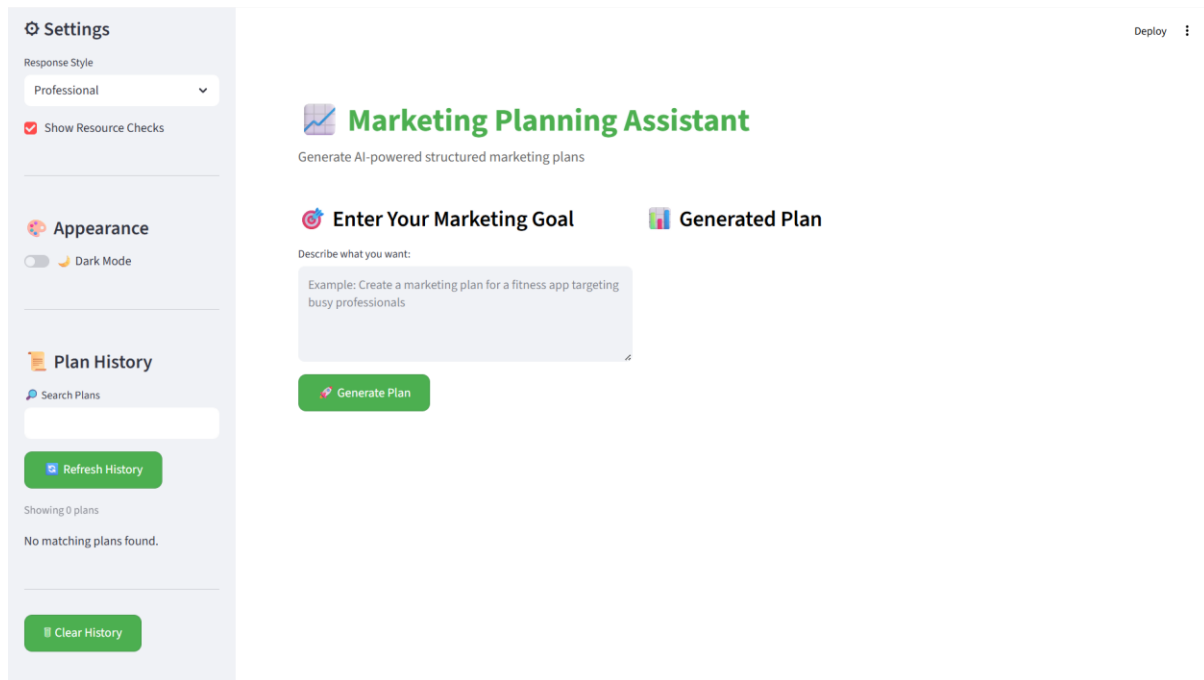
#### Development Environment:-

- **PyCharm IDE** – Used for coding, debugging, and project management

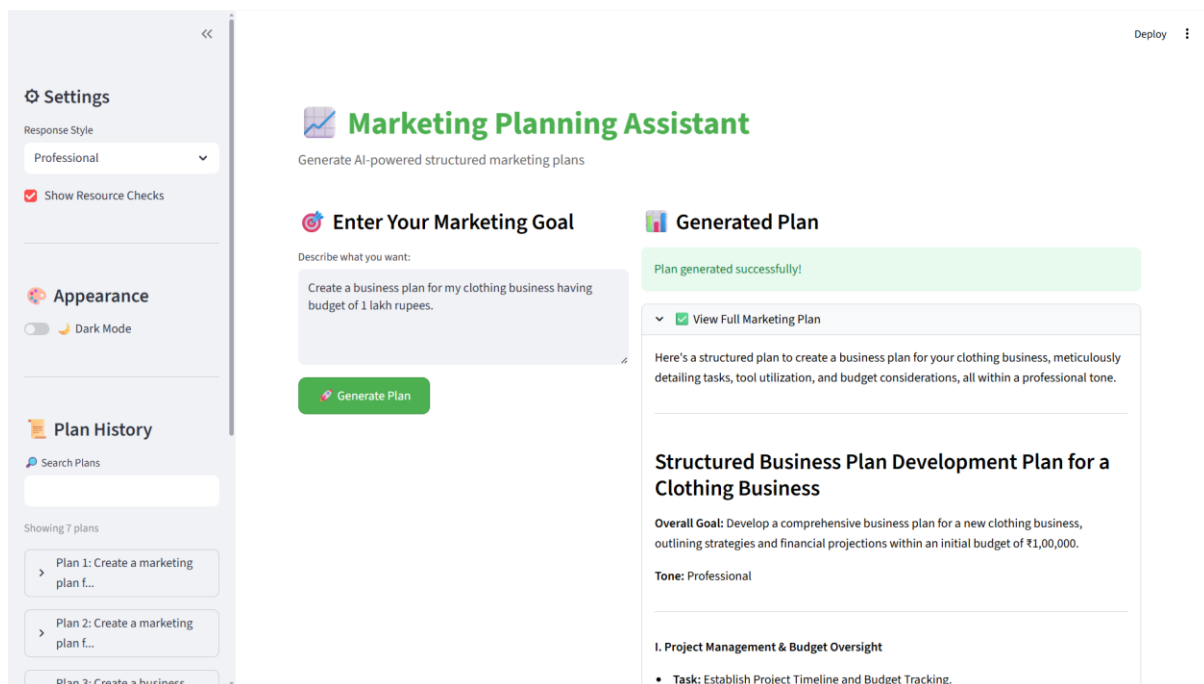
# Results & Output

## 1. Screenshots / outputs

### ➤ Main UI

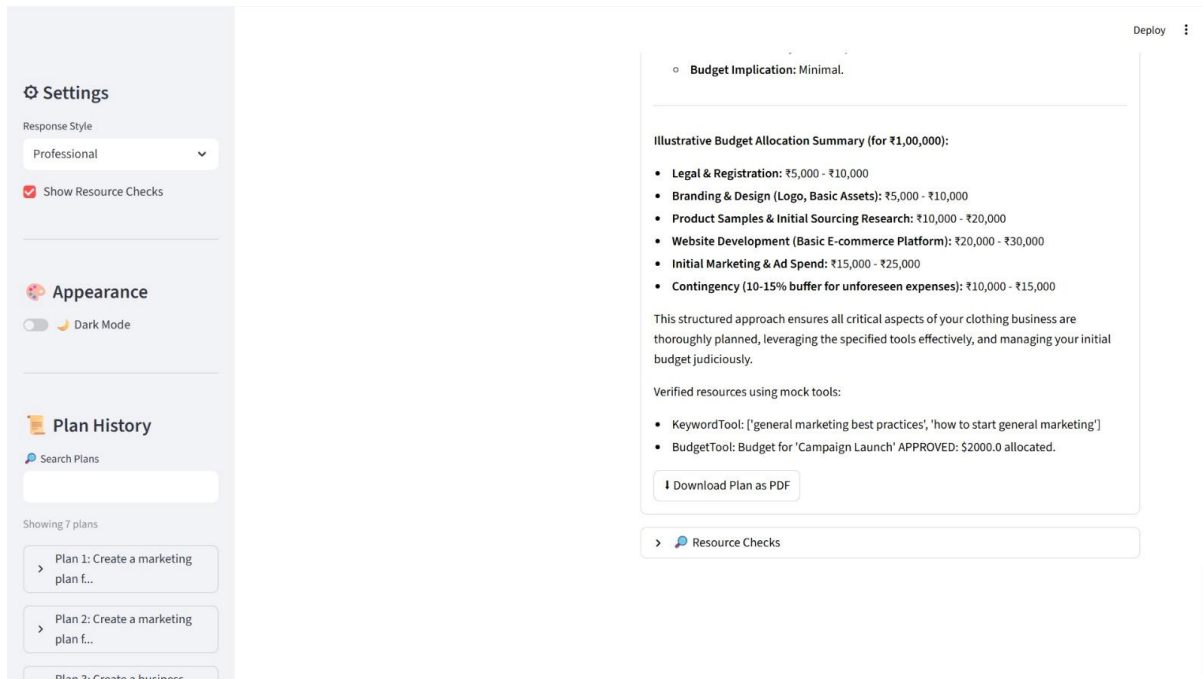


### ➤ Goal Input & Plan Generation

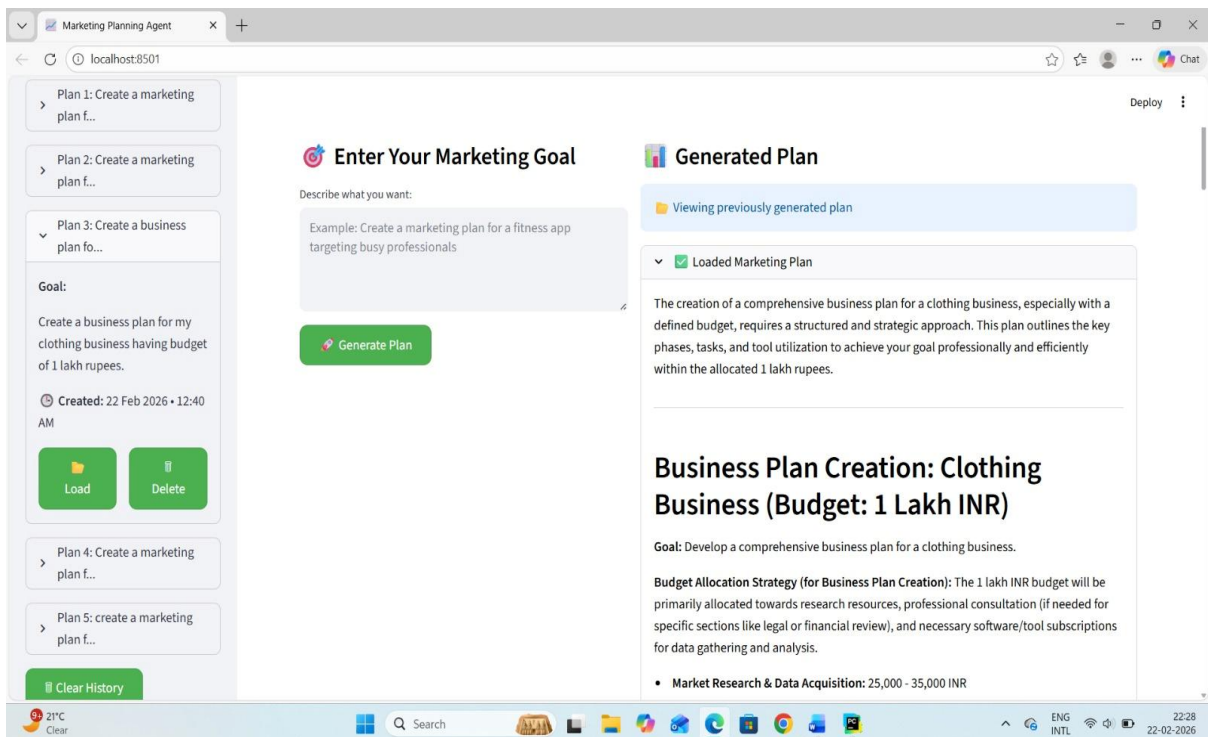




## ➤ PDF Export Feature



## ➤ Load Previous Plan



## ➤ Search/Filter Plans

☒ Show Resource Checks

**Appearance**  
☐ Dark Mode

**Plan History**  
  
  
 Showing 3 plans  

Plan 1: Create a marketing plan f...

Plan 2: Create a marketing plan f...

Plan 3: Create a marketing plan f...

Clear History

**Marketing Planning Assistant**  
 Generate AI-powered structured marketing plans

**Enter Your Marketing Goal**

**Generated Plan**

Describe what you want:  

Create a business plan for my clothing business having budget of 1 lakh rupees.

Generate Plan

## ➤ Database Records(MySQL Workbench) & Design

MANAGEMENT  
 Server Status  
 Client Connections  
 Users and Privileges  
 Status and System Variables  
 Data Export  
 Data Import/Restore  
 INSTANCE  
 Startup / Shutdown  
 Server Logs  
 Options File  
 PERFORMANCE  
 Dashboard  
 Performance Reports  
 Performance Schema Setup

SQL File 4  

```

1 CREATE DATABASE marketing_agent;
2 USE marketing_agent;
3 CREATE TABLE plans (
4   id INT AUTO_INCREMENT PRIMARY KEY,
5   goal TEXT NOT NULL,
6   plan TEXT NOT NULL,
7   created_at DATETIME DEFAULT CURRENT_TIMESTAMP
8 );
9
10 SHOW TABLES;
11 select * from plans;

```

SQLAdditions  
 Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

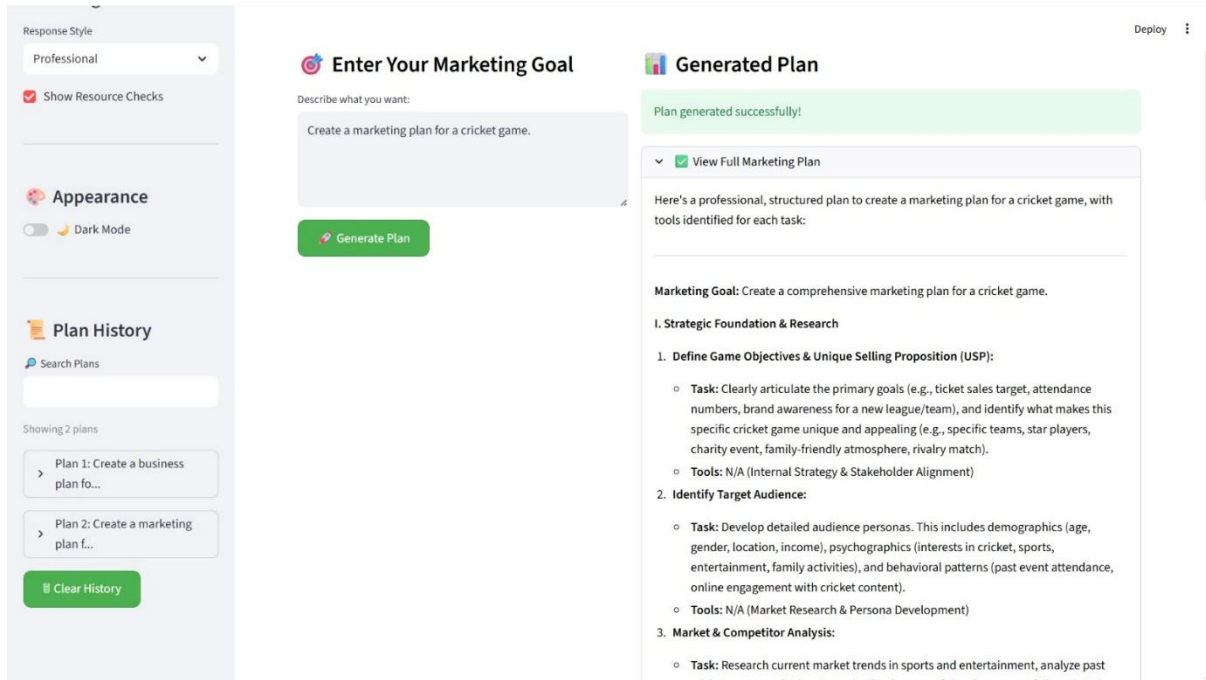
| id | goal                                                | plan                                                 | created_at          |
|----|-----------------------------------------------------|------------------------------------------------------|---------------------|
| 1  | Create a marketing plan for a fitness app.          | Here's a professional decomposition of creating ...  | 2026-02-22 00:14:48 |
| 2  | Create a marketing plan for a cricket game          | As an expert Marketing Planning Assistant, I will... | 2026-02-22 00:17:08 |
| 4  | Create a business plan for my clothing business ... | The creation of a comprehensive business plan ...    | 2026-02-22 00:40:02 |
| 5  | Create a marketing plan for fitness app targetin... | Here's a structured marketing plan for a fitness ... | 2026-02-22 00:57:04 |
| 12 | create a marketing plan for fitness app targetin... | As an expert Marketing Planning Assistant, I will... | 2026-02-22 21:57:29 |

plans 1 x  
 Output  
 Action Output  

| # | Time     | Action                            | Message           | Duration / Fetch      |
|---|----------|-----------------------------------|-------------------|-----------------------|
| 1 | 21:58:15 | USE marketing_agent               | 0 row(s) affected | 0.000 sec             |
| 2 | 21:58:19 | select * from plans LIMIT 0, 1000 | 5 row(s) returned | 0.000 sec / 0.000 sec |

## 2. Reports / Dashboards / Models

### ➤ AI-Generated Marketing Plan(output content)



**Response Style**  
 Professional  
☒ Show Resource Checks

**Appearance**  
☐ Dark Mode

**Plan History**  
 Search Plans  
 Showing 2 plans  
 Plan 1: Create a business plan fo...  
 Plan 2: Create a marketing plan L...  
 Clear History

**Enter Your Marketing Goal**  
 Describe what you want:  
 Create a marketing plan for a cricket game.  
 Generate Plan

**Generated Plan**  
 Plan generated successfully!  
☒ View Full Marketing Plan  
 Here's a professional, structured plan to create a marketing plan for a cricket game, with tools identified for each task:

**Marketing Goal:** Create a comprehensive marketing plan for a cricket game.

**I. Strategic Foundation & Research**

**1. Define Game Objectives & Unique Selling Proposition (USP):**

- Task:** Clearly articulate the primary goals (e.g., ticket sales target, attendance numbers, brand awareness for a new league/team), and identify what makes this specific cricket game unique and appealing (e.g., specific teams, star players, charity event, family-friendly atmosphere, rivalry match).
- Tools:** N/A (Internal Strategy & Stakeholder Alignment)

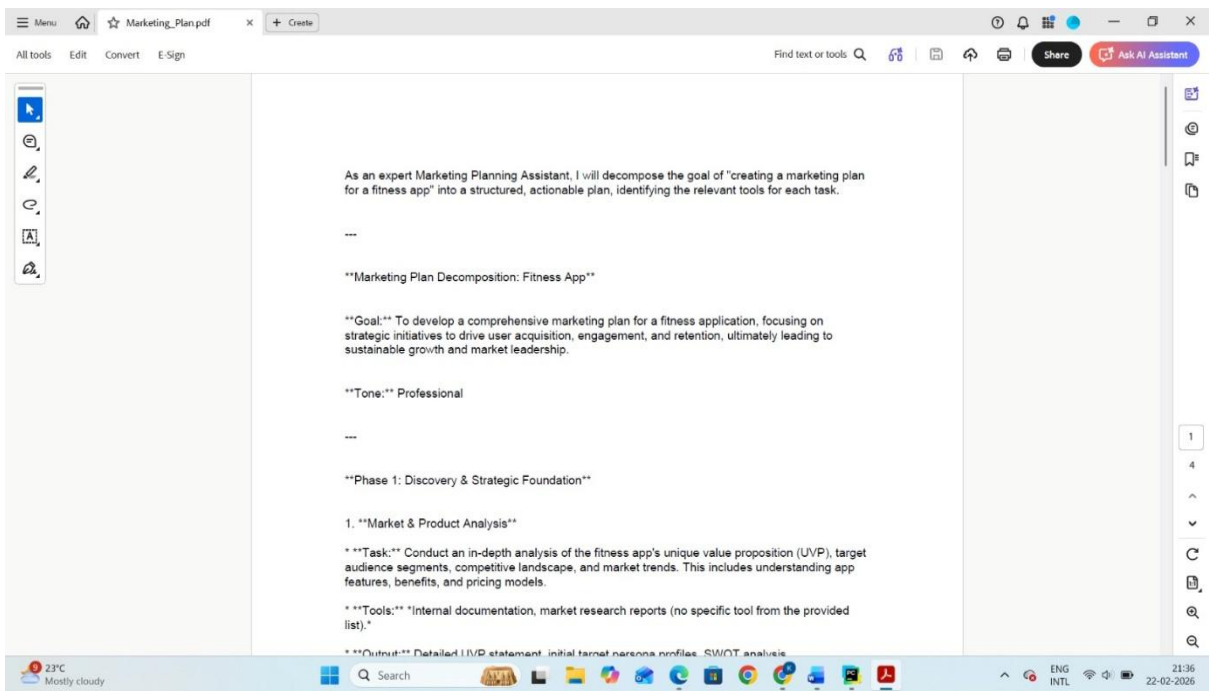
**2. Identify Target Audience:**

- Task:** Develop detailed audience personas. This includes demographics (age, gender, location, income), psychographics (interests in cricket, sports, entertainment, family activities), and behavioral patterns (past event attendance, online engagement with cricket content).
- Tools:** N/A (Market Research & Persona Development)

**3. Market & Competitor Analysis:**

- Task:** Research current market trends in sports and entertainment, analyze past

### ➤ Exported PDF Marketing Plan (Report Output)



As an expert Marketing Planning Assistant, I will decompose the goal of "creating a marketing plan for a fitness app" into a structured, actionable plan, identifying the relevant tools for each task.

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**\*\*Marketing Plan Decomposition: Fitness App\*\***

**\*\*Goal:\*\*** To develop a comprehensive marketing plan for a fitness application, focusing on strategic initiatives to drive user acquisition, engagement, and retention, ultimately leading to sustainable growth and market leadership.

**\*\*Tone:\*\*** Professional

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**\*\*Phase 1: Discovery & Strategic Foundation\*\***

**1. \*\*Market & Product Analysis\*\***

**\*\*Task:\*\*** Conduct an in-depth analysis of the fitness app's unique value proposition (UVP), target audience segments, competitive landscape, and market trends. This includes understanding app features, benefits, and pricing models.

**\*\*Tools:\*\*** Internal documentation, market research reports (no specific tool from the provided list).

**\*\*Output:\*\*** Detailed UVP statement, initial target persona profiles, SWOT analysis.

### 3. Key outcomes

- Functional AI-powered marketing planning assistant developed.
- Automated structured marketing plan generation achieved.
- End-to-end workflow from user input to output implemented.
- Integration of Large Language Model using LangChain completed.
- Interactive GUI-based interface using Streamlit designed.
- PDF export and downloadable reporting implemented.
- Persistent MySQL database storage integrated.
- Enhanced UX features including history, search, deletion, and dark mode added.

## Conclusion

The AI-Powered Marketing Planning Assistant Agent effectively demonstrates the practical application of Artificial Intelligence, Large Language Models (LLMs), LangChain, and Streamlit in addressing real-world marketing strategy challenges. The system successfully automates the generation of structured and actionable marketing plans based on high-level user goals.

By integrating LangChain with Google Generative AI, the agent provides intelligent reasoning and strategic recommendations. The Streamlit-based graphical interface enhances usability through intuitive interaction, while features such as plan history, search, deletion, dark mode, and PDF export improve overall user experience.

Additionally, the implementation of MySQL database ensures persistent storage and efficient management of generated plans. This project not only achieved its intended functional objectives but also provided valuable hands-on experience in AI integration, prompt engineering, UI design, database connectivity, and debugging.

## Future Scope & Enhancements

- Multi-user login system
- Cloud deployment
- Advanced analytics dashboard
- Real API integrations (Google Ads, Meta Ads)
- Campaign performance tracking
- AI-based budget optimization
- Task execution automation